



## SagarDrishti

*A browser-native 3D digital twin of India's ocean*

- **Problem Statement ID** - SIH26067
- **Problem Statement Title** - Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations.
- **Theme** - Disaster Management
- **PS Category** - Software
- **Team ID** - 158120
- **Team Name** - PixelPaws



Running today, offline: **24** depth levels from 5 m to 2000 m, the **26°C** isosurface, **337** current vectors and CAP warning areas.

**Not a proposal on paper.** 607 automated tests · 51 fps at 1080p on integrated graphics · 10.5 M gridded values · 30,012 QC-passed instrument levels · zero requests that leave the machine. Every figure in this deck was measured, not estimated.

# The Problem & Our Solution

Bay of Bengal: model fields and the instruments in the water, on one screen, with a number for the gap between them.

## THE PROBLEM · INCOIS'S OWN WORDS

- **“No integrated, web-based 3D visualization platform currently exists”** so forecasters toggle between disparate software packages, which impedes hazard assessment, search and rescue, fishery advisories and climate monitoring.

## PROPOSED SOLUTION · HOW IT ADDRESSES THE PROBLEM

**Volumetric column in a browser.** 24 levels, 5 m to 2000 m.

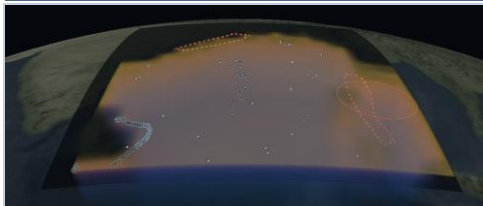
**All four instrument classes.** 149 casts: temperature, salinity, oxygen, chlorophyll.

**A skill number beside the field.** Bias and RMSE per depth band.<sup>[1]</sup>

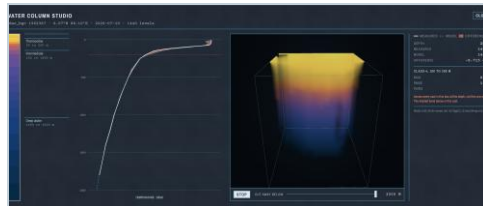
**Hazards drawn over the water.** Live CAP v1.2 from NDMA SACHET.<sup>[20][15]</sup>

**A new source is config, not code.** YAML plus a plugin. OGC WMS/WCS served.<sup>[18][19]</sup>

## THE RUNNING SYSTEM · PHOTOGRAPHED TODAY

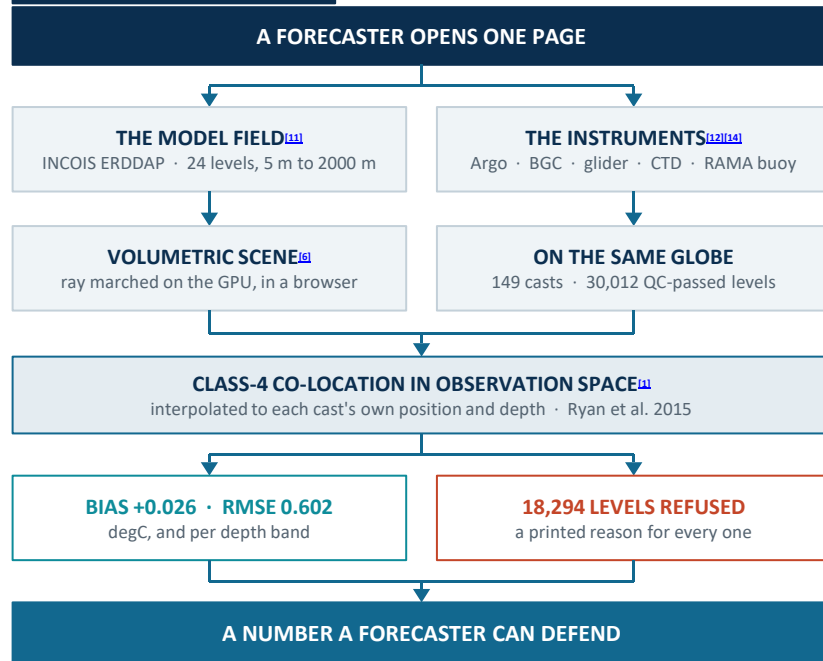


One scene: the model volume, the instruments, a live CAP area



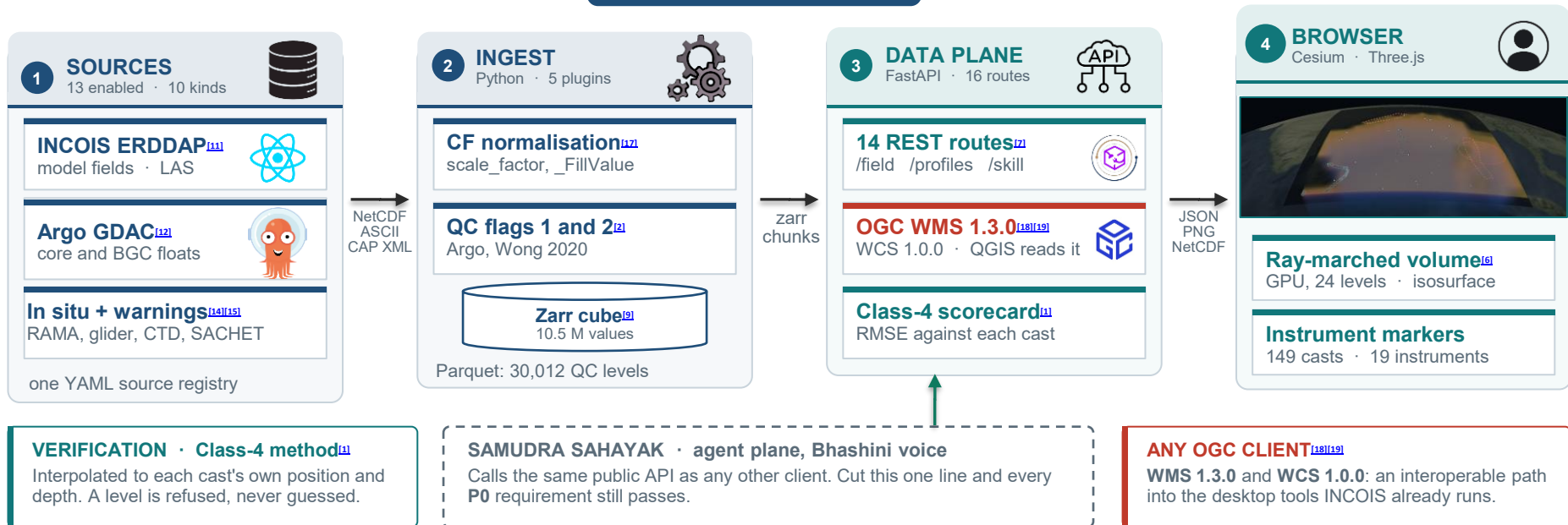
The cube, the measured profile, and the Class-4 number

## HOW ONE ANSWER GETS MADE





## SYSTEM ARCHITECTURE



## TECHNOLOGIES USED



Standards: CF Conventions · OGC WMS 1.3.0 · OGC WCS 1.0.0 · OASIS CAP v1.2 · TEOS-10 · Argo QC (Wong 2020) · Class 4 (Ryan 2015)



FEASIBILITY IS NOT A FORECAST HERE. THE SYSTEM ALREADY RUNS. EVERY FIGURE BELOW WAS MEASURED, NOT ESTIMATED

607

automated tests, 12 in a browser

51 fps

at 1080p on integrated graphics

0

requests leave the machine

10.5 M

gridded values, two agencies

30,012

QC-passed levels, 19 instruments

12

ocean parameters, 3 by plugins

## ANALYSIS OF FEASIBILITY

- **Proven stack.** CesiumJS, FastAPI, Zarr, Docker.
- **Frame budget cleared.** 51 fps on the weakest GPU.
- **Reads INCOIS's own ERDDAP.** No duplication.
- **Zero licence cost.** One GPU server, open source.



### THE SAME BUILD, ON A PHONE

- 1 variable selector
- 2 colorbar, min / max, log
- 3 24 levels, 5 m to 2000 m
- 4 vertical exaggeration
- 5 layer opacity
- 6 time-step animation

## POTENTIAL CHALLENGES AND RISKS

### DATA

- **Instrument coverage is uneven.** Not on INCOIS ERDDAP.
- **Datasets disagree on axes.** GLORYS stops at 1942 m.

### TECHNICAL

- **Volumetric rendering is costly.** Translucent overdraw dominates.
- **Scaling to the full basin.** The national grid is larger.

### ADOPTION AND TRUST

- **A wrong number beats no number.** Its caveat must travel.
- **A false alarm is unacceptable.** Drills are dashed.

## STRATEGIES TO OVERCOME THEM

### DATA

- **Parsers already written.** A file, not new code.
- **Never stretch one grid onto another.** The served depth is printed.

### TECHNICAL

- **Depth-space level of detail.** At most 10 of 24 slices at once.<sup>[6]</sup>
- **Chunked Zarr over range requests.** Cost is view-dependent.<sup>[9]</sup>

### ADOPTION AND TRUST

- **The caveat travels with the number.** On screen and in the API.
- **Four independent drill guards.** Exercise is refused.

## VIABILITY · WHY THIS SURVIVES AFTER THE HACKATHON

### Native deployment path

docker compose up, 12 browser tests pass.

### Standards, not a silo

WMS 1.3.0 and WCS 1.0.0: their tools read it.

### Maintainable by the owner

Standard Python and JavaScript.

### Costs nothing to keep

Zero licences, one GPU server.



## POTENTIAL IMPACT ON THE TARGET AUDIENCE

**DISASTER MANAGERS**

A tsunami or high-wave CAP bulletin becomes a shape on the water.

**OPERATIONAL FORECASTERS**

"Can I trust today's field?" answered per depth band.

**SEARCH AND RESCUE**

Depth-resolved currents, which is what a drifting object follows.

**FISHERY ADVISORIES**

Subsurface temperature behind advisories reaching ~9 lakh households.

**RESEARCH, EDUCATION, OUTREACH**

Guided tours narrate the ocean for schools and exhibitions.

## BENEFITS · SOCIAL, ECONOMIC, ENVIRONMENTAL

- **Faster hazard assessment.** One tool replaces the toggling the PS describes; the model and the instruments are already side by side.
- **Verifiable, not decorative.** Bias +0.026 degC and RMSE 0.602 degC over 11,718 matched pairs, with the analysis-fit caveat attached.
- **Trust where it matters most.** The model is within about 0.12 degC below 500 m and 2.07 degC across the thermocline at 50 to 100 m, which is exactly the layer a cyclone forecaster reads.

- **Zero licence cost.** Fully open source against proprietary 3D suites; one deployment serves the nation.
- **Resilient by design.** Runs air-gapped, so a network failure at a control room or a nodal centre does not take the tool down.
- **Wider access.** A browser and, in the next phase, Indian-language voice widen who can interrogate the ocean beyond specialists.

## NATIONAL ALIGNMENT

Digital India

Deep Ocean Mission

Blue Economy

SAGAR / MoES mandate

UN SDG 14: Life Below Water

Deployable on INCOIS infrastructure

The PS is an internal INCOIS tooling request, so the deployment story is native rather than argued.

## WHAT CHANGES ON THE DAY IT MATTERS · A DEPRESSION FORMS IN THE BAY

**1 SEE THE HEAT**

Switch to the depth of the 26°C isotherm, the layer a cyclone feeds on.

**2 CHECK THE MODEL**

The card says the field is 2.07°C off across 50 to 100 m.

**3 CONFIRM AGAINST THE WATER**

Click the nearest Argo float. Its measured profile against the model.

**4 PUT THE WARNING IN CONTEXT**

The CAP bulletin is drawn over the water it concerns. Network off.

## DETAILS AND LINKS OF THE REFERENCE AND RESEARCH WORK

## METHOD AND VERIFICATION

- [1] [Ryan et al. 2015: GODAE OceanView Class 4 verification, J. Op. Oceanogr. 8](#) · the method behind our scorecard  
 [2] [Wong et al. 2020: Argo data and its QC flags, Front. Mar. Sci. 7:700](#) · why we keep only flags 1 and 2  
 [3] [IOC, SCOR, IAPSO: TEOS-10 thermodynamic equation of seawater](#) · our density field  
 [4] [Balakrishnan Nair et al. 2013: INCOIS Ocean State Forecast validation, Current Science 105\(2\)](#) · the Indian validation precedent

## WEB 3D OCEAN VISUALISATION

- [5] [Qin et al. 2021: Web 3D framework for time-varying ocean forecast data, Env. Modelling & Software 135](#) · the closest ancestor  
 [6] [Yu, Qin & Xu 2025: WebGPU volume rendering of ocean scalars, Applied Sciences 15\(5\):2782](#) · our rendering optimisations  
 [7] [Tucker et al. 2020: Argovis, J. Atmos. Ocean. Tech. 37\(3\):401](#) · the API contract we follow  
 [8] [Blower et al. 2013: ncWMS, Env. Modelling & Software 47](#) · OGC WMS over CF-NetCDF  
 [9] [Signell & Pothina 2019: Cloud coastal model analysis, JMSE 7\(4\):110](#) · why Zarr rather than whole NetCDF files  
 [10] [pyParaOcean \(IIsc\): desktop ocean volumetrics we bring to the browser](#)

## DATA SOURCES WE INGEST

- [11] [INCOIS ERDDAP: the analysis field, 24 levels, 5 to 2000 m](#)  
 [12] [Argo GDAC \(Ifremer\): Indian Ocean profile files](#)  
 [13] [Copernicus Marine: GLOBAL\\_ANALYSISFORECAST\\_PHY\\_001\\_024 currents](#)  
 [14] [NOAA PMEL: RAMA moored buoy array, Indian Ocean](#)  
 [15] [NDMA SACHET: India's national CAP alert backbone](#)  
 [16] [INCOIS: Indian National Centre for Ocean Information Services](#)

## OPEN STANDARDS IMPLEMENTED

- [17] [CF Conventions for climate and forecast metadata](#)  
 [18] [OGC Web Map Service \(WMS\) 1.3.0](#)  
 [19] [OGC Web Coverage Service \(WCS\) 1.0.0](#)  
 [20] [OASIS Common Alerting Protocol \(CAP\) v1.2](#)

## HOW THE RESEARCH CHANGED THE BUILD · not a bibliography, a set of decisions

Ryan 2015 → verify in observation space<sup>[1]</sup>

Wong 2020 → QC flags 1 and 2 only<sup>[2]</sup>

Yu 2025 → drop a slice that cannot change the picture<sup>[3]</sup>

Signell 2019 → Zarr chunks over HTTP range reads<sup>[4]</sup>

## WHERE WE SIT · GAP ANALYSIS

| Capability                           | Desktop tools | Web viewers | Digital Ocean | Sagar Drishti |
|--------------------------------------|---------------|-------------|---------------|---------------|
| Browser-native, no install           | ✗             | ✓           | ✓             | ✓             |
| True volumetric water column         | ✓             | ✗           | ~             | ✓             |
| Model + in-situ in ONE 3D scene      | ✗             | ✗           | ~             | ✓             |
| Model-vs-observation skill on screen | ✗             | ✗           | ✗             | ✓             |
| CAP hazard layer in ocean context    | ✗             | ✗           | ✗             | ✓             |
| Runs fully offline / air-gapped      | ✓             | ✗           | ✗             | ✓             |

Desktop tools: ODV, IDV, pyParaOcean. Web viewers: Argovis, MyOcean. INCOIS Digital Ocean portal advertises 3D and 4D; what it does not do is fuse instruments into the volume or publish model skill beside the field.

## OUR OWN WORK, AND HOW TO CHECK IT

**607** automated tests, 12 in a real browser

**6 dp** TEOS-10 check values reproduced exactly

**100%** of panels cite dataset and timestamp

**15 + 8** papers and existing tools reviewed for the gap analysis